The Compute Merit Order: A Negative Result

Electricity cost is roughly one per cent of GPU rental revenue, and the geographic curtailment sequence it implies is not binding

Compute Merit Order Project

Working paper — draft, August 4, 2026

Abstract

A GPU cluster earns a compute price per GPU-hour and pays an electricity cost per GPU-hour. The difference is a spread superficially analogous to the spark spread in power markets, and it implies a shutdown price below which a fleet is uneconomic to run. Because regional power costs differ, shutdown prices differ, which is said to imply a global merit order in which compute capacity curtails in a predictable geographic sequence. We construct that merit order from open data for seven European bidding zones and find that **it exists, is computable, and is not binding**. Power ranges from 1.0 % of revenue at H100 on-demand list prices to 17.4 % at A100 spot prices, so the answer depends critically on which revenue basis is used — a choice the spark-spread framing obscures and which moves the result by more than any parameter in the cost model. Headroom over avoidable cost is at least 76 % on every basis and in every zone we test, no zone is uneconomic anywhere in the grid of assumptions, and compute prices would have to fall a further 76 % to 99 % before the first zone curtails. The dispersion in shutdown prices across our sample is 1.53x, not the 3x the thesis assumes, and adding non-power operating cost compresses it further toward 1.17x. We also find the supply-rich/constrained hub taxonomy does not predict cost in this sample: the cheapest zone is France, nominally constrained, and Norway NO2, nominally supply-rich, is the second dearest. Separately, we recover Azure's 1, 3 and 5-year committed rates from published reservation data — a 60 % discount to list that is routinely described as confidential — and use them as the revenue basis a committed operator actually faces. The spark-spread analogy fails quantitatively because fuel is 60 % to 90 % of a gas plant's marginal cost while power is single-digit percentages of a GPU fleet's revenue on any realistic basis. We state the conditions under which the thesis would become true, and note that the interesting economics lies in capital recovery and siting, not in shutdown.

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1 Introduction

Spark spreads — the difference between the wholesale price of electricity and the fuel cost of generating it — are a standard tool in power-market economics, used to decide when a gas-fired plant is worth dispatching. An analogous quantity is often asserted to govern AI compute: a GPU cluster sells compute at some price per GPU-hour, its main variable cost is electricity, and when the compute price falls below that cost plus non-power operating expense, a rational operator curtails. Because industrial power costs vary substantially across regions, the shutdown prices implied by this logic vary too, which would produce a global merit order — a predictable geographic sequence in which compute capacity curtails as compute prices fall.

This paper tests that proposition and reports a negative result. The mechanism is real and we compute it explicitly, but its magnitude is far too small for it to bind at observed prices. The reason is a simple quantitative asymmetry that the analogy obscures: for a combined-cycle gas turbine, fuel is typically 60 % to 90 % of short-run marginal cost, so a change in gas prices moves the dispatch decision directly. For a GPU fleet renting at on-demand list prices, electricity is of order 1 % of revenue, rising to 17 % only at the cheapest spot price we observe. A threefold difference in a one-per-cent cost item is a two-percentage-point difference in margin, not a dispatch signal.

We test two claims:

1. **The compute merit order.** That regional power-cost differences produce a geographic curtailment sequence as compute prices fall (Section 5). We test curtailment, not retirement: the two are distinct decisions with distinct cost bases, and conflating them is a category error we discuss in Section 3.

2. **Chips versus power.** That the binding constraint on AI buildout is grid interconnection and power availability rather than chip supply (Section 6).

We are able to test the first and report it as not binding. We are *not* able to test the second with the data assembled so far, and we say so rather than gesture at it (Section 6).

A note on timing. We do not claim, as earlier drafts of this project did, that comparable compute-price data first appeared in 2026: at least one commercial index has published weekly observations since July 2024 (`docs/SOURCES.md`), and our own sources file contradicted that earlier claim. What is new here is not the existence of such data but the construction of a *freely redistributable* series from vendor-published prices, which is what makes an independently auditable spread computable at all.

2 Data and methods

2.1 Free-data constraint

This project publishes its dataset under CC-BY-4.0 and holds itself to a hard constraint: **a source is eligible only if it is free to access and permits redistribution of derived values**. A number a reader cannot fetch and re-derive is not auditable, and a number we are contractually barred from republishing cannot enter `/data/final` at all. This ruled out the two credible commercial compute-price indices (Section 8) and, as described below, five of the twelve power zones we collected.

2.2 Power prices

Day-ahead prices come from the Energy-Charts API operated by Fraunhofer ISE (Fraunhofer ISE, 2026), which requires no authentication. We collected 2026 year-to-date hourly prices for twelve European bidding zones. Energy-Charts returns a licence *per zone*, and they differ: seven zones arrive as CC-BY-4.0 via Bundesnetzagentur/SMARD, while five exchange-sourced zones (Spain, Portugal, Finland, Sweden SE3, Italy North) carry "for private and internal use only". Those five are fetched and analysable locally but their derived values are withheld from the published dataset; the omission is recorded in `docs/WITHHELD_ZONES.md` rather than left silent. All results below therefore rest on seven zones, which is a materially smaller sample than we set out to use.

Prices are converted from EUR/MWh to USD kW^{-1} h at the ECB monthly-average reference rate (European Central Bank, 2026), mean 1.163 365 over the period.

2.3 Compute prices

Because no free redistributable compute-price index exists (Section 8), we construct one from published cloud list prices, which are facts about public offers and can be recorded with attribution. We query the Azure Retail Prices API (Microsoft, 2026b), which is free and unauthenticated, for a fixed basket of GPU SKUs across three regions. Accelerator counts per SKU — required to convert a price per instance-hour into a price per GPU-hour — are read from Microsoft's public documentation repository (Microsoft, 2026a) and recorded per SKU with their own source URL, never inferred from SKU naming.

We filter to Linux rows. Windows rows are excluded because they embed OS licensing, which is not a cost of running an accelerator, and expired price rows are excluded because they are history.

We publish five price bases: on-demand list, spot, and 1, 3 and 5-year reserved. The reserved series matters. Committed-rate pricing is widely described — including in earlier drafts of this paper — as confidential and unobtainable. It is not: Azure publishes reservation totals through the same free API, and dividing by the term's hours yields a comparable hourly rate. The 5-year H100 rate is 60 % below list. Any analysis using list price as the revenue of a committed operator overstates it by roughly 2.5x, and we previously made exactly that error.

Accelerator	Price basis	USD per GPU-hour	% of list
H100	On-demand list	14.13	100
H100	Reserved 1-year	9.05	64
H100	Reserved 3-year	6.20	44
H100	Reserved 5-year	5.65	40
H100	Spot	2.61	18
A100	On-demand list	4.32	100
A100	Spot	0.82	19

Table 1: Published Azure price bases, mean across sampled regions. The choice of basis moves the revenue leg by 5x and is therefore a larger modelling decision than anything in the cost model.

2.4 Cost model

Delivered power per accelerator-hour is

$$\text{delivered\_kW} = (\text{TDP} + \text{overhead}) \times \text{PUE} = (0.700 + 0.180) \times 1.25 = 1.10\,\text{kW}, \tag{1}$$

held as a single named constant and swept over 0.9 kW to 1.4 kW in Section 7. We flag that the 0.180 kW overhead term is a residual chosen to reconcile the other two figures with a 1.10 kW total, not an independent measurement.

3 The compute spark spread

Power cost per GPU-hour is

$$\text{power\_cost} = \text{delivered\_kW} \times \text{price}_{\text{USD/kWh}}. \tag{2}$$

The shutdown price adds non-power marginal operating expense:

$$\text{shutdown\_price} = \text{power\_cost} + \text{opex}_{\text{marginal}}. \tag{3}$$

Headroom against a compute price is

$$\text{headroom} = \frac{\text{compute\_price} - \text{shutdown\_price}}{\text{compute\_price}}. \tag{4}$$

What this cost is, and is not. For a fleet whose accelerators are already purchased, capital cost is sunk and therefore irrelevant to the decision to keep running: the operator curtails when revenue falls below *avoidable* cost. Equation (3) is that avoidable cost, and it is the correct quantity for a shutdown question. It is emphatically not the right quantity for an investment question. A fleet can sit far above its shutdown price while being a value-destroying investment, because capital recovery dominates its full cost. We compute only the former, and Section 9 argues the latter is where the economically interesting question actually lives.

We do not assign a point estimate to $\text{opex}_{\text{marginal}}$. No public source for it was identified, so it is swept over 0.00 USD to 0.20 USD per GPU-hour and every result below is reported across that sweep.

4 Regional cost surfaces

4.1 Power

Across the seven publishable zones, 2026 year-to-date mean day-ahead prices run from 0.0847 USD $\mathrm{kW}^{-1}\,\mathrm{h}$ (France) to 0.1296 USD $\mathrm{kW}^{-1}\,\mathrm{h}$ (Austria), a spread of 1.53x. Negative price hours are common, totalling 1823 across the sample, which matters because it means the power cost of compute is occasionally negative and the shutdown constraint cannot bind at all in those hours.

The hub taxonomy does not predict cost. The project's own supply-rich versus constrained classification fails on this sample. The cheapest zone is France, classified constrained. Norway NO2, classified supply-rich, is the second dearest at 0.1142 USD $\mathrm{kW}^{-1}\,\mathrm{h}$ — it is the southern Norwegian zone, heavily interconnected to Germany and Great Britain, and therefore prices toward its neighbours rather than toward Nordic hydro. Any hub-level narrative that assumes "Nordics are cheap" without checking the specific bidding zone is wrong, and the geographic intuition behind the merit-order thesis is weaker than it appears.

4.2 Compute

As of 2026-08-03, Azure's published on-demand list price averages 14.13 USD per GPU-hour for H100 and 4.32 USD per GPU-hour for A100 across the regions sampled (Figure 1). Spot prices for the same silicon clear at roughly a fifth of list, so the dispersion *within* the compute price exceeds the dispersion in power costs that motivates this paper.

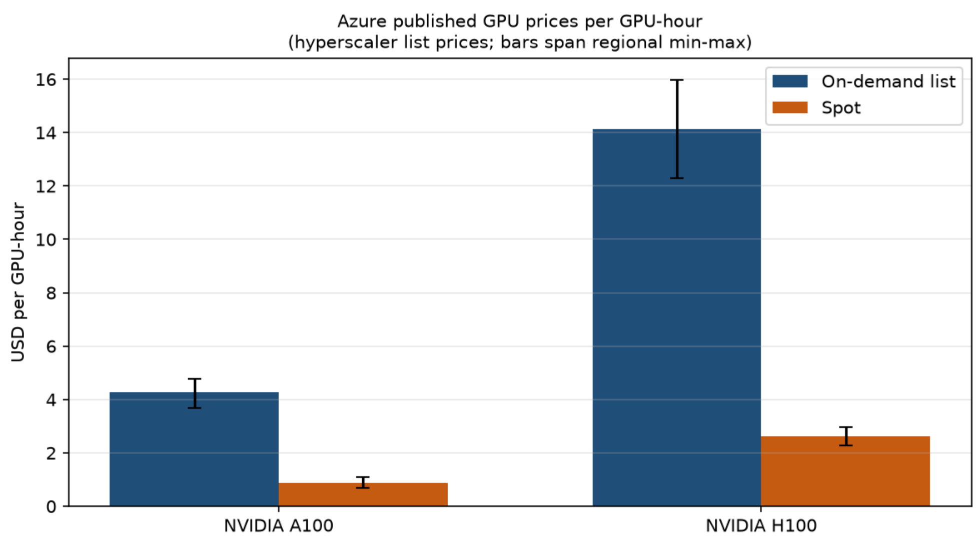

Figure 1: Published Azure GPU prices per GPU-hour. Bars span the regional minimum to maximum. On-demand list price is an upper bound on realised revenue; spot is shown for contrast, not as a revenue proxy.

5 The merit order

Figure 2 is the central result. Panel (a) is the merit order itself: zones sorted ascending by avoidable cost per GPU-hour. Panel (b) places that ladder on the same logarithmic axis as observed compute prices, which is the only way to show both on one chart — and the gap between them is the finding.

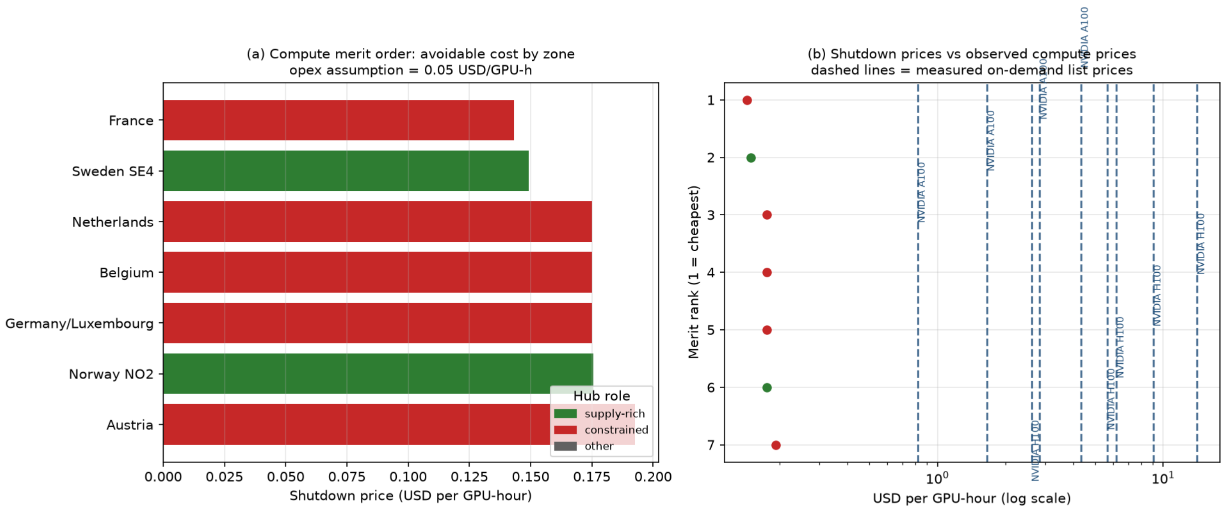

Figure 2: The compute merit order. (a) Avoidable cost by zone at the midpoint opex assumption. (b) The same shutdown prices against measured on-demand compute prices, log scale. The ordering is real; the gap is two orders of magnitude.

5.1 The order exists

The retirement sequence is well defined. At the midpoint opex assumption and H100 prices, zones would go uneconomic in the order Austria, Norway NO2, Germany/Luxembourg, Belgium, Netherlands, Sweden SE4, France. So the qualitative claim — that a geographic sequence exists and is computable from public data — survives.

5.2 But it is not binding

Electricity is 1.01 % of revenue in the dearest zone at H100 on-demand list prices, and 17.4 % at A100 spot prices. That range is the most important number in this paper: the spark-spread analogy is hopeless at the top of it and merely weak at the bottom, and which end applies is decided entirely by the revenue basis, not by anything about electricity.

Minimum headroom over avoidable cost is 98.6 % at H100 on-demand list and 76.4 % at A100 spot, and *no zone is uneconomic under any combination of price basis, zone and opex we test.* Compute prices would have to fall a further 76 % to 99 % before the first zone — Austria — curtails. A merit order whose first event requires the cheapest observed price to fall by three quarters again is not a description of any plausible near-term market.

5.3 The dispersion is smaller than assumed

The thesis assumes roughly 3x variation. We measure 1.53x at zero opex, falling to 1.17x at 0.20 USD per GPU-hour.

This compression is not an artefact of our sample; it is algebraic. If power costs span $[p, 3p]$ and non-power opex is a common constant c, the shutdown-price ratio is

$$\frac{3p+c}{p+c}, \tag{5}$$

which equals 3 only when $c = 0$ and decreases monotonically in c. The headline "3x power costs imply 3x shutdown prices" inference is therefore false for any non-zero non-power operating cost, independently of data. Since plausible non-power opex is of the same order as — or larger than — the entire power cost, the dispersion that actually drives a merit order is materially below the dispersion in electricity prices that motivates it.

5.4 Why this cannot be rescued by better power data

The gap is not a measurement problem. Two comparisons from our own data make the scale clear. Azure's list price for the same H100 SKU differs by 3.69 USD per GPU-hour between `eastus` and `westeurope` — a vendor pricing decision. The entire power-cost difference across our seven zones is 0.05 USD per GPU-hour. The revenue leg's regional dispersion is roughly 70x the cost leg's, and it is driven by commercial policy rather than by electricity.

Capital cost sharpens the same point. An 8-accelerator H100 node at 300 kUSD to 400 kUSD over four to five years is on the order of 1 USD to 1.7 USD per GPU-hour of depreciation before any cost of capital — roughly 10 to 50x the power cost. We do not publish these figures in our dataset because we have no free, citable source for node prices, and they are stated here only to size the comparison. But the ordering is unambiguous: for any question other than short-run curtailment, power is not the term that matters.

6 Constraint analysis: chips versus power

[Pending: not tested. Interconnection-queue and grid-headroom data collection has not been carried out, so this paper offers no evidence on whether power or chip supply binds. We note only that the finding above is consistent with power mattering for siting and capital planning *rather than for dispatch, since a cost that is one per cent of revenue can still be decisive when the alternative is not being able to energise a site at all. That is a different mechanism from the merit order and requires different data.]*

7 Sensitivity

The conclusion is robust to every parameter we vary, because the result is not marginal — it is two orders of magnitude from the threshold.

The honest way to state the fragility is this: the conclusion depends far more on the *level* of the compute price than on anything in the cost model. If the relevant compute price is not the hyperscaler list price but something closer to the spot price — roughly a fifth of list — headroom falls from 99 % to about 95 %. That is still not binding. Only a compute price near 0.15 USD per GPU-hour, some 95x below current list, would make the merit order operative.

8 Limitations

We list every material weakness rather than the flattering subset.

Parameter	Range tested	Effect on headroom	Changes conclusion?
Non-power opex	0.00 USD to 0.20 USD/GPU-h	99.0 % → 97.6 %	No
Accelerator	A100 vs H100	96.7 % → 99.0 %	No
Zone	7 European zones	1.53 x cost spread	No
delivered_kW	0.9 kW to 1.4 kW	±27 % on a 1 % item	No

Table 2: Sensitivity. To overturn the conclusion, compute prices would have to fall roughly 40-fold, or power costs rise roughly 40-fold; no combination of the parameters above comes close.

- **Seven zones, one continent, one partial year.** Licence restrictions removed five of twelve collected zones from the published sample, including Iberia and the Nordic zones that would most likely widen dispersion. Our 1.53x is therefore a lower bound on global dispersion, and the US, Gulf, China and India hubs are absent entirely.
- **No capacity weighting.** A merit-order curve should have installed capacity on its x-axis. We have no per-zone GPU capacity data, so Figure 2 shows the cost ladder without capacity weights. The retirement *sequence* is unaffected; the quantity retiring at each step is unknown.
- **The compute-price leg is the weakest input.** No free redistributable index exists: Silicon Data's `SDH100RT` and GetDeploying's index are both paywalled, and the latter's terms prohibit redistributing raw data. We rejected both under the free-data constraint rather than cite figures a reader cannot verify, and built our own basket instead — which is narrower than what we rejected.
- **One vendor, one segment.** The basket is Azure-only and hyperscaler-only. No neocloud provider is represented, so the segment dispersion we argue is central cannot yet be measured within a consistent methodology.
- **On-demand list price is not realised revenue.** Large operators transact on confidential multi-year reserved contracts at material discounts. This is the mirror of the PPA problem on the power side: both legs are public proxies for confidential transactions. Using list price *overstates* revenue and therefore overstates headroom.
- **Wholesale power price is not delivered power cost.** We compare a hyperscaler retail compute price against a wholesale commodity power price. A real operator pays network charges, levies and taxes on top of the wholesale price, which in several European jurisdictions exceed the commodity component. This *understates* power cost, possibly by more than a factor of two, and is the single most consequential methodological gap in this paper. Correcting it would raise the power share of revenue but would not plausibly close a hundredfold gap.
- **PUE and overhead are assumptions, not measurements.** Operator-reported PUE is self-reported and unaudited, and our 0.180 kW overhead is a reconciling residual.
- **Capital cost is excluded by construction.** This is correct for a curtailment question and wrong for every other question one might ask. Our own rough sizing puts depreciation at 10 to 50x the power cost.
- **All power cost is treated as avoidable.** That is wrong for any operator on an industrial tariff with a demand charge, or under a take-or-pay PPA, where a large part of the power bill

is incurred whether or not the machine runs. Correctly splitting avoidable from unavoidable power cost would *lower* the true shutdown price and make the merit order bind even less often than we report.

- **Claims 1 and 2 are in tension and we cannot resolve it.** If interconnection is genuinely the binding constraint, then grid capacity is contracted and largely pre-paid, and an operator holding a scarce interconnect runs it regardless of hourly energy price — which would suppress the very curtailment mechanism Claim 1 describes. The two claims can only both hold under specific contracting regimes, and we have not established which regime prevails.

- **Azure spot is not an auction clearing price.** In our data spot sits at a near-constant 18.5 % of list across two accelerator generations and two continents, which is the signature of a published discount schedule rather than a market. We use it as a low revenue bound, not as evidence about where capacity clears.

- **List prices are stale relative to power prices.** Several price rows carry effective dates years old, while the power series is hourly. Any spread between a sticky annual list price and an hourly spot power price is dominated by the staleness of the former.

9 Conclusion

The compute merit order is real, computable from open data, and economically inert at current prices. Electricity is about one per cent of H100 rental revenue; headroom over avoidable cost exceeds 97 % everywhere we can measure; no zone is uneconomic; and the cross-zone dispersion in shutdown prices is 1.53x rather than the 3x assumed, compressing further once non-power operating costs are included.

The spark-spread analogy fails quantitatively. It works for gas generation because fuel is most of marginal cost, so fuel-price differences drive dispatch. It does not transfer to GPU compute, where power is a rounding error against rental revenue. Regional power-price differences of the magnitude observed cannot produce a retirement sequence; they can only produce small margin differences.

This does not make power irrelevant to AI infrastructure — it relocates where power matters. Three candidate mechanisms survive our result and each requires different data than we assembled:

1. **Capital recovery, not dispatch.** If accelerator depreciation dominates full cost, the economically interesting threshold is the price at which new capacity stops being *built*, not the price at which existing capacity stops running. Our shutdown price is deliberately the wrong tool for that question.

2. **Availability, not price.** A one-per-cent cost item is still decisive when the alternative is being unable to energise a site. Interconnection queues and grid headroom plausibly bind on *whether* capacity exists rather than on what it costs to run.

3. **Compute price collapse.** If compute prices fell toward marginal cost, as they would in a commoditised market with excess capacity, the merit order would become operative. Our result quantifies how far that is: roughly a 40- to 95-fold decline.

We report this as a negative result because the data does not support the thesis as framed. The project's next phase should redirect toward constraint data and capital cost, where the evidence suggests the binding economics actually lies.

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